

Enmanuel Leon

Senior Fullstack & Backend Engineer | AI Tooling · Distributed Systems · Cloud (AWS/GCP)

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Professional Summary

Computer Engineer with 6+ years of experience in mission-critical software engineering, specializing in distributed architectures, AI workflow automation, and transactional platforms. Proven track record designing high-availability asynchronous services with **TypeScript, Node.js (Fastify/Express), and Python** deployed across cloud infrastructure (**AWS / GCP**). Expert in architecting high-throughput document processing engines, secure M2M orchestration for isolated VPC environments, and developing payment gateway integrations across heterogeneous API ecosystems (REST, GraphQL, SOAP) backed by relational databases (MySQL, PostgreSQL) and high-concurrency message queues (**Redis, BullMQ, SQS**).

Technical Skills

- **Backend & Architecture:** Node.js (Fastify, Express), Python (FastAPI), Microservices, REST, GraphQL, SOAP, Event-Driven Architecture (EDA), M2M Communication, PHP/Lumen service maintenance
- **Cloud, Networking & DevOps:** AWS (ECS, Lambda, RDS, S3, SQS, CloudWatch, IAM), GCP (Cloud Run), Docker, CI/CD (GitHub Actions), NGINX, Linux, VPC Isolation & Networking
- **Databases & Messaging:** MySQL, PostgreSQL, MongoDB, Redis, BullMQ, AWS SQS, Row-Level Security (RLS)
- **Applied AI & Automation:** LLM Integrations (OpenAI, Anthropic APIs), Asynchronous Artifact Generation, Agent Orchestration, RAG, Prompt Engineering, Tool Use
- **Frontend & Mobile:** React, Next.js, Vite, React Native (Cross-platform iOS/Android), TypeScript, JavaScript (ES6+), HTML5, Tailwind CSS
- **Testing, Quality & Observability:** Vitest, Jest, Datadog, Sentry, CloudWatch, TDD

Professional Experience

Fullstack Developer (AI & Core Systems) | Entropy Systems | Barcelona, Spain (Remote)

April 2024 - Present

- **Reduced report generation processing time by 65%** by designing and deploying a distributed asynchronous document and artifact generation engine using Fastify and Python, processing over 2,000 complex audit reports monthly via BullMQ and Redis queues.
- **Architected the backend layer of the core AI tooling platform** (agents, workflows, and skills), delivering high-concurrency multi-instance horizontal scaling and low-latency (<50ms) Machine-to-Machine (M2M) communication via decoupled APIs and token-based authentication.
- **Designed a periodic asynchronous synchronization engine** using distributed crons and background workers that securely interconnects isolated customer VPC environments with the central core platform, maintaining 99.9% data consistency without exposing public endpoints.
- **Operated distributed microservices across AWS and GCP with 99.9% uptime**, containerizing services in Docker and implementing CI/CD pipelines that reduced deployment times from ~1 hour to ~10 minutes, backed by real-time observability in Sentry and CloudWatch.

Software & Mobile Engineer (Freelance Consulting) | Fintech Sector (NDA)

2024 - 2025

- **Engineered an end-to-end cross-platform financial mobile solution** (iOS and Android) using React Native and TypeScript, spearheading technical requirements definition, system architecture, and cloud service integrations.
- **Architected scalable cloud data infrastructure and security controls**, implementing robust authentication flows and fine-grained Row-Level Security (RLS) policies to protect sensitive transactional records.
- **Optimized mobile application performance and network resilience**, reducing load times by 35% and securing reliable offline state persistence for transactional operations in low-connectivity environments.

Fullstack Developer | Interfell - ePayco | Remote

June 2021 - April 2024

- **Maintained 99.9% service uptime** across financial payment gateway services processing over 500,000 monthly transactions by engineering and maintaining transactional microservices in Node.js (Express) and TypeScript on MySQL, while integrating and supporting auxiliary services in PHP (Lumen).
- **Led technical payment gateway integrations** for corporate merchants and partners, orchestrating transaction lifecycles between responsive React web applications and backend services connected across REST, GraphQL, and legacy SOAP APIs.
- **Reduced payment processing latency by 28%** by optimizing critical MySQL queries, offloading asynchronous background tasks with AWS SQS queues, and refactoring transactional API orchestration layers.
- **Decreased Mean Time to Resolution (MTTR) by 40%** by establishing reproducible Docker environments, comprehensive automated testing suites in Jest, and proactive anomaly monitoring via Datadog and Sentry.

Systems Analyst / Software Developer | Intelix Synergy, C.A. | Valencia, Venezuela

March 2020 - June 2021

- **Boosted business process efficiency by 30%** by delivering 3 modular internal management systems in JavaScript, Node.js, and Python used daily by over 200 active employees.
- **Ensured continuous zero-unplanned-downtime availability** by administering production Linux servers configured with NGINX reverse proxies, load balancing, and SSL/TLS security policies.
- **Decreased developer onboarding time by 50%** by authoring comprehensive OpenAPI/Swagger specifications, interactive endpoint documentation, and modular architecture blueprints.

Education

Bachelor of Science in Computer Engineering

Universidad José Antonio Páez | Valencia, Venezuela

2017 - 2020

Additional Information

- **Languages:** Spanish (Native), English (Technical / Conversational B1 - Fluent technical documentation, tickets, architectural design docs, and async written collaboration in distributed teams).
- **Areas of Specialization:** Distributed Systems Architecture, Multi-Cloud (AWS · GCP), Applied AI Automation, Payment Gateways & Fintech Integrations.
- **Portfolio & Code:** enmanuel23x.github.io | github.com/enmanuel23x